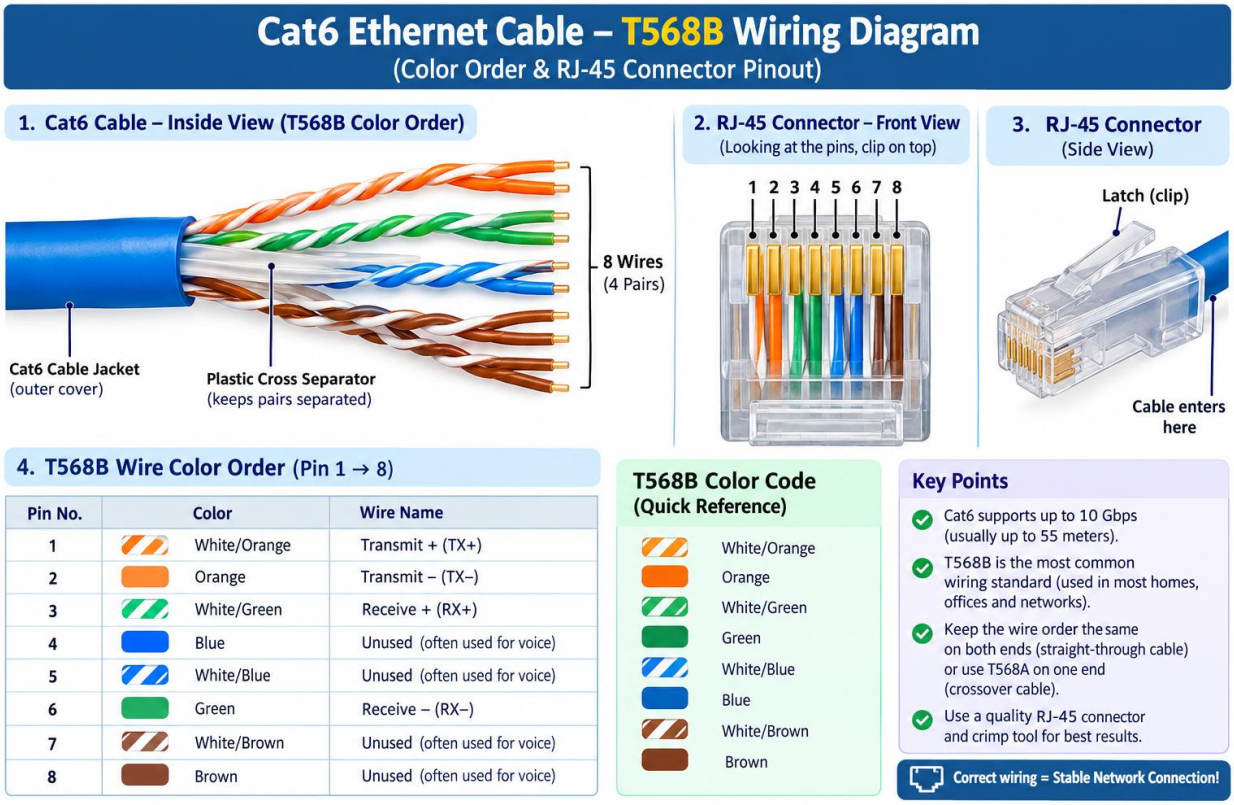
1. **Draw a labeled diagram of a Cat6 Ethernet cable, showing the color order of wires in the T568B standard and labeling the RJ-45 connector pins.**



2. **List three types of network cables (such as Cat5e, Cat6, and fiber optic) and for each, mention one real-world app or device (like a WiFi router, Zomato POS terminal, or IPL streaming server) where it would be used, explaining why.**

Ans. 1. Cat5e Cable: Used to connect a Wi-Fi router to a computer because it provides a reliable internet connection for everyday use.

2. Cat6 Cable: Used to connect a Zomato POS terminal in a restaurant because it supports fast data transfer and helps process orders smoothly.

3. Fiber Optic Cable: Used to connect an IPL streaming server because it transfers large amounts of data at very high speeds over long distances, helping deliver live matches smoothly.

3.Watch a YouTube video showing how to crimp an RJ-45 connector onto a network cable, then write down the step-by-step process in your own words and mention two common mistakes to avoid.

Ans. Step-by-step process

1. Cut the cable: Cut the Ethernet cable to the required length using a cable cutter.

2. Remove the outer cover: Carefully strip about 2–3 cm of the outer jacket to expose the eight small wires inside.

3. Straighten the wires: Separate the four twisted pairs and straighten all eight wires gently.

4. Arrange the wires: Follow the T568B colour order:

White/Orange → Orange → White/Green → Blue → White/Blue → Green → White/Brown → Brown.

5. Trim the wires: Cut all eight wires evenly so they can fit properly inside the RJ-45 connector.

6. Insert the wires: Push all eight wires fully into the connector, keeping them in the correct order. Make sure the cable's outer jacket also reaches inside the connector.

7. Crimp the connector: Place the connector into the crimping tool and press firmly to secure the wires.

8. Test the cable: Use a LAN cable tester to check that all eight connections are working correctly.

Two common mistakes to avoid

1. Wrong wire colour order: If the wires are arranged incorrectly, the network cable may not work properly. Always check the T568B colour sequence.

2. Wires not inserted fully: If the wires do not reach the connector contacts, the connection may fail. Push them fully inside before crimping.

Conclusion: Correct wire order, proper crimping, and testing help make a reliable Ethernet cable.

4.Given the following scenario: You are setting up a gaming zone for an IPL screening event and need to connect multiple screens and consoles. Choose the appropriate cable and connector types for high-speed, reliable connections, and explain your choices.

Ans. For setting up a gaming zone for an IPL screening event, I would choose the following cables and connectors for fast and reliable connections.

1. Cat6 Ethernet Cable with RJ-45 Connector: I would use this to connect gaming consoles and smart TVs to the network router because it provides a fast, stable internet connection with less interference.

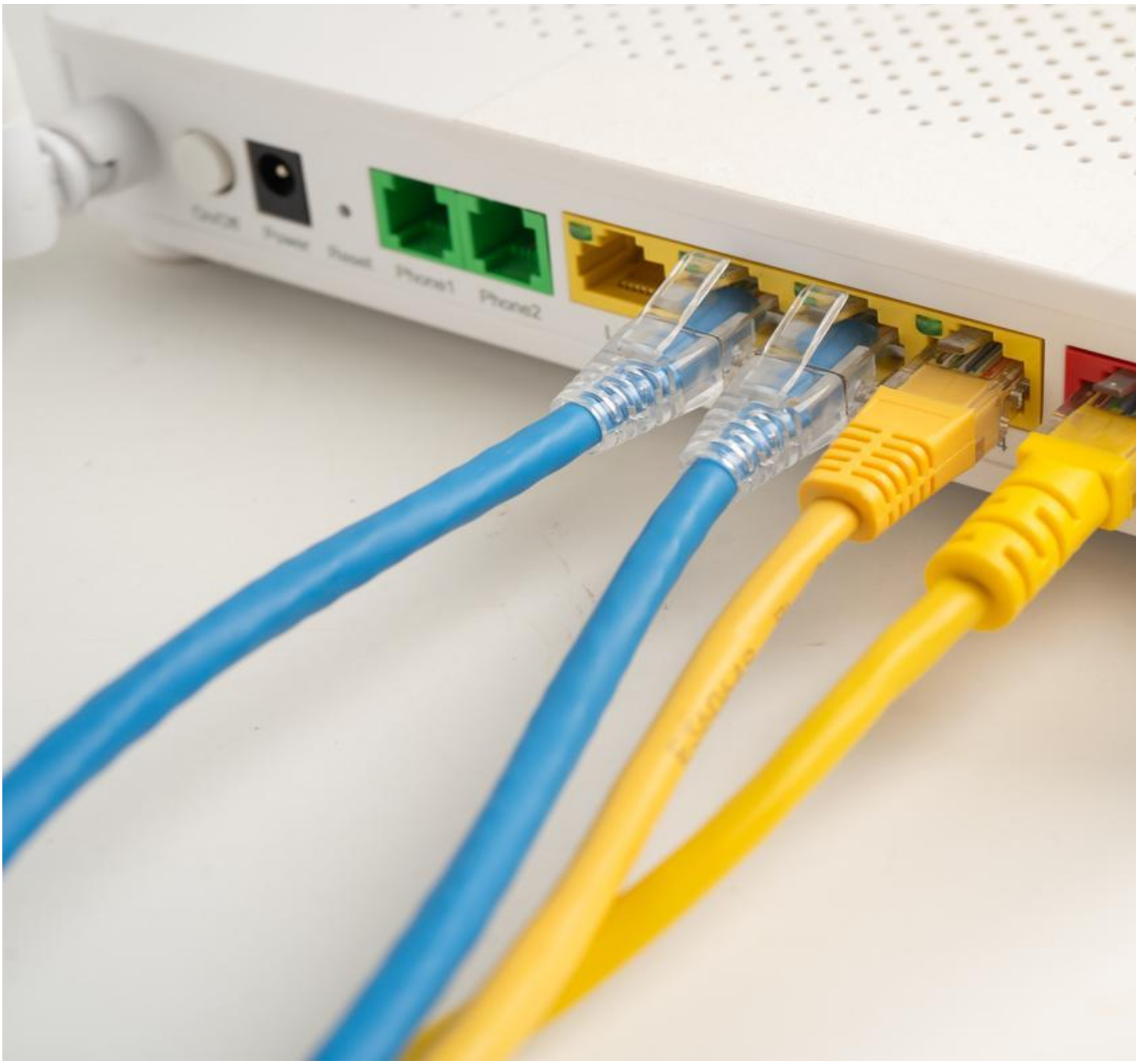
2. HDMI Cable: I would use this to connect gaming consoles to TVs or large screens because it carries high-quality video and audio with low delay.

3. Fiber Optic Cable: I would use this to connect the main network to different areas of a large gaming zone because it supports high-speed data transfer over long distances.

Conclusion: Cat6 with RJ-45 is best for network connections, HDMI is best for connecting consoles to screens, and fiber optic is useful for high-speed connections across a large venue.

**5.Find and share a photo (from your home, college, or online) of any network cable or connector in use (such as behind a WiFi router or at a café's POS counter), and describe what type it is and what devices it connects.

Hint: Look for blue or yellow cables with clear plastic ends (RJ-45) or thin round cables for fiber optics.**



- Cable type: Ethernet cable (Cat5e or Cat6).
- Connector type: RJ-45 connector — the clear plastic plug at the end of the cable.
- Devices connected: The cable connects a Wi-Fi router to a computer, switch, or another network device.
- Purpose: It transfers data between devices and provides a stable, wired network connection.

Conclusion: I identified the cable as an Ethernet cable with an RJ-45 connector. It is used to connect network devices and provide reliable internet access.